
Toward Auditable Vision-Language Control

A Fully Tensor-Decomposable VLA

Specialist capability and exact weight-level diagnostics

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Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ -VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A mechanical audit finds no hidden incompatible operation, and local reconstructions agree near the float32 precision floor. On all ten LIBERO-Object tasks, the convolutional 20.1M-parameter instantiation reaches $93.7\% \pm 0.8\%$ under the published three-seed protocol. Averaging its three fixed checkpoints reaches 96.2% at three times the inference cost. In a separate three-checkpoint ViT instantiation, we derive an exact weight-based decomposition of softmax-free attention and reconstruct every attention module with worst relative error 2.56×10^{-7} . All 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. The evidence establishes strong specialist capability together with exact, checkpoint-native access to learned attention structure. Such access is relevant when deployment failures may involve visual shortcuts or weak language use. The present results do not establish broad robotic generalization, robust instruction grounding, selective behavioral editing, or an exact end-to-end decomposition of the full policy.

1 Introduction

Deployable vision-language robot policies need more than aggregate success. When a policy fails, developers need to distinguish possible visual shortcuts, weak instruction use, state-estimation errors, and unstable action generation. Conventional checkpoints expose little of this structure directly. Post-hoc probes can identify correlated activations, but their conclusions depend on a chosen dataset and need not reveal how the deployed weights construct an action.

Tensor decomposition offers a complementary, checkpoint-native analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. This structure does not certify semantic grounding by itself. It provides a direct weight-level surface for subsequent grounding and failure analyses.

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. Exact structural access is useful only if the restricted policy still controls reliably.

We therefore study a prerequisite for deployment-oriented diagnostics:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while exposing exact layerwise structure from its weights?

We test capability and structural access in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit. These experiments test whether a useful diagnostic surface can coexist with control, not whether that surface already certifies grounding or supports a validated edit.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. Our narrower result is that an algebraic VLA can make exact layerwise weight structure directly accessible without giving up strong specialist control.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^T) \odot (Q_2 K_2^T)/s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

78 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 79 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

80 Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a
 81 numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization
 82 operations update this pair. The model therefore keeps input-specific rescaling while remaining
 83 exactly representable as a rational tensor network.

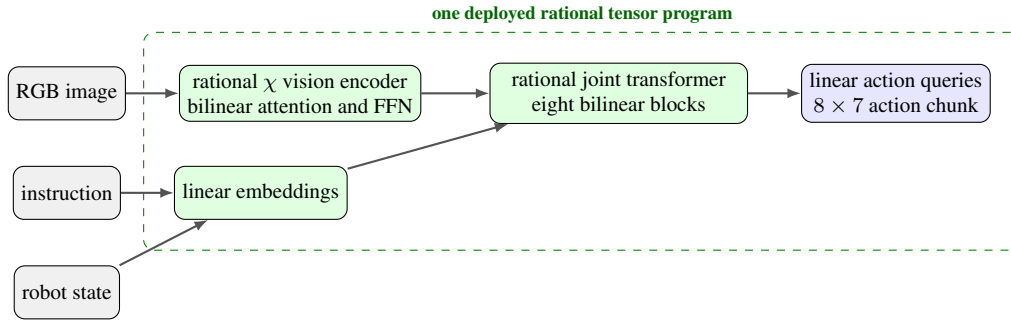


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

84 2.2 Architecture and training

85 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 86 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 87 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 88 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 89 continuous action dimensions. The complete model has 20,137,352 learned parameters.

90 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 91 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 92 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 93 that verify every requested state was loaded.

94 2.3 Mechanical certificate on the trained checkpoint

95 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 96 point at runtime and reconstruct representative operations from saved weights.

Mechanical decomposability audit of the trained 189/200 checkpoint

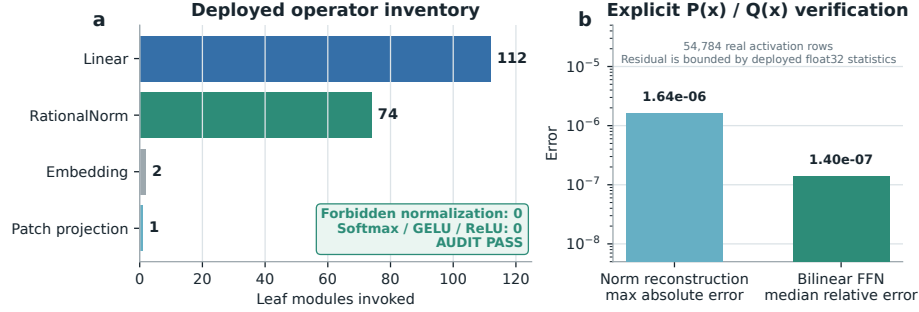


Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

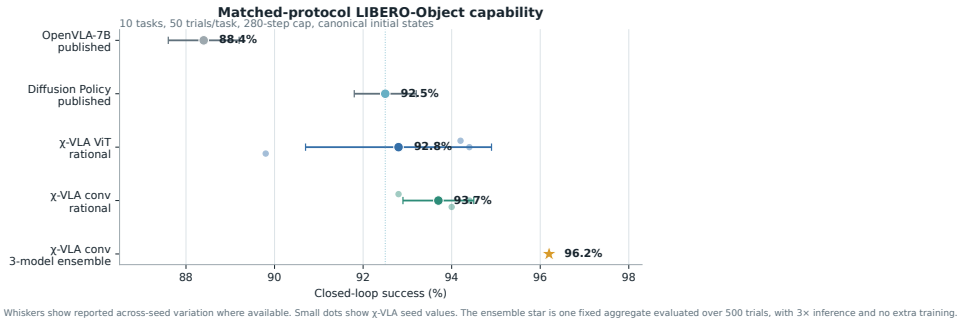


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are X-VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

110 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

111 The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high
 112 specialist capability is reproduced across three independently trained checkpoints. These are historical
 113 Modal aggregate records. A locked replay is producing immutable episode-level outcomes.

114 Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and
 115 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points
 116 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato
 117 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one
 118 20.1M-parameter policy.

119 3.3 What this result does and does not establish

120 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
 121 existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad
 122 robotic generalization or state of the art.

123 4 Exact weight-derived structure through attention

124 4.1 Exact layerwise construction

125 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
 126 a high-order coefficient table. Each entry says how strongly one combination of query, key, and
 127 value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe
 128 are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and
 129 an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to
 130 1.24×10^{-14} .

131 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 132 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 133 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 134 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 135 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
 136 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
 137 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
 138 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

139 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 140 values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays
 141 the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most
 142 coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force
 143 materialization to 2.13×10^{-14} .

144 4.2 Trained policy result

145 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
 146 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
 147 sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace.

148 Real cached activations are used only afterward to measure how faithfully that subspace preserves
 149 the layer’s computation.

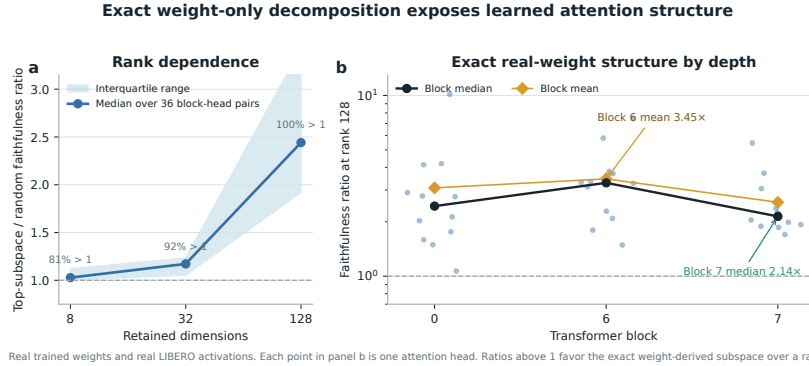


Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

150 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
 151 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
 152 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
 153 0, 6, and 7.

154 This extends exact weight-only construction through individual attention layers. It is not an exact
 155 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
 156 rapid degree and representation growth.

157 5 Related work

158 **VLA capability.** OpenVLA established an open pretrained VLA and standardized LIBERO com-
 159 parisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can
 160 substantially improve adaptation and control efficiency [2]. These systems use large pretrained
 161 backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

162 **Tensor and polynomial networks.** Polynomial networks and tensor networks use explicit mul-
 163 tilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight
 164 eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for com-
 165 positional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an
 166 exact reduced construction for individual softmax-free attention layers.

167 **Causal mechanisms and policy analysis.** Bilinear IMPALA policies have combined competitive
 168 ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5].
 169 This is the closest architectural precedent for weight-based policy analysis. Recent work steers
 170 pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify
 171 and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions
 172 recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These
 173 results make a universal separation from conventional policies untenable. Our intended distinction is
 174 empirical: exact weight auditing in a strong continuous vision-language policy.

175 **VLA robustness.** Recent evaluations show that standard LIBERO success can coexist with weak
 176 robustness and language use [9–11]. This motivates model-internal diagnostics, but our exact weight
 177 analysis does not itself certify grounding. We therefore treat LIBERO-Object as a bounded specialist
 178 capability setting rather than evidence of broad deployment readiness.

6 Limitations and conclusion

χ -VLA demonstrates that strong specialist closed-loop control can coexist with a mechanically auditable rational graph and exact layerwise attention analysis. This combination is relevant to deployment diagnostics because it exposes trained coefficient structure without fitting a separate probe.

The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions. The experiments do not show that the policy follows language when visual cues conflict, generalizes across task distributions, or transfers to physical robots. Exact reconstruction is certified layer by layer and does not yet yield one compact symbolic contraction of the full policy. The three-checkpoint ensemble trades three forward passes for its strongest capability result.

We therefore present χ -VLA as diagnostic infrastructure, not as a deployment-ready or robustly grounded robot policy. Convolutional checkpoints support the capability result, while separate ViT checkpoints support the exact reconstruction. Broader suites, counterfactual instruction tests, and physical-robot evaluation remain necessary for deployment claims. We do not claim a direct coefficient-edit interface.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

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A Supporting results and decisions

A.1 Per-task ensemble behavior

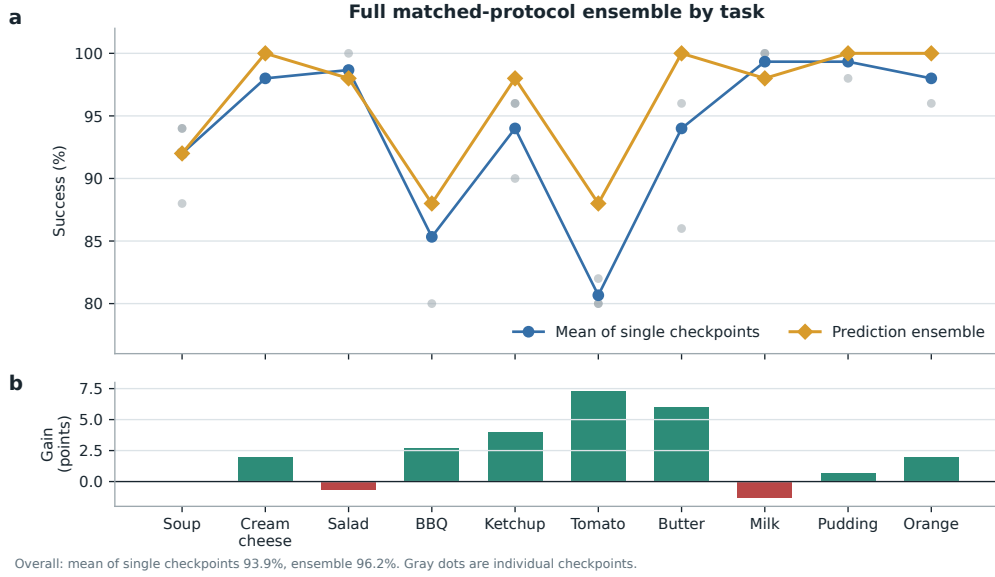


Figure 5: Per-task effect of prediction averaging in the paired ensemble evaluation. Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.2 Expanded attention screen

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.3 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.